

Unit IV Activity II: R Codes, Console Outputs & Plot Outputs

Problem 1: Retail Daily Sales Time Series Analysis

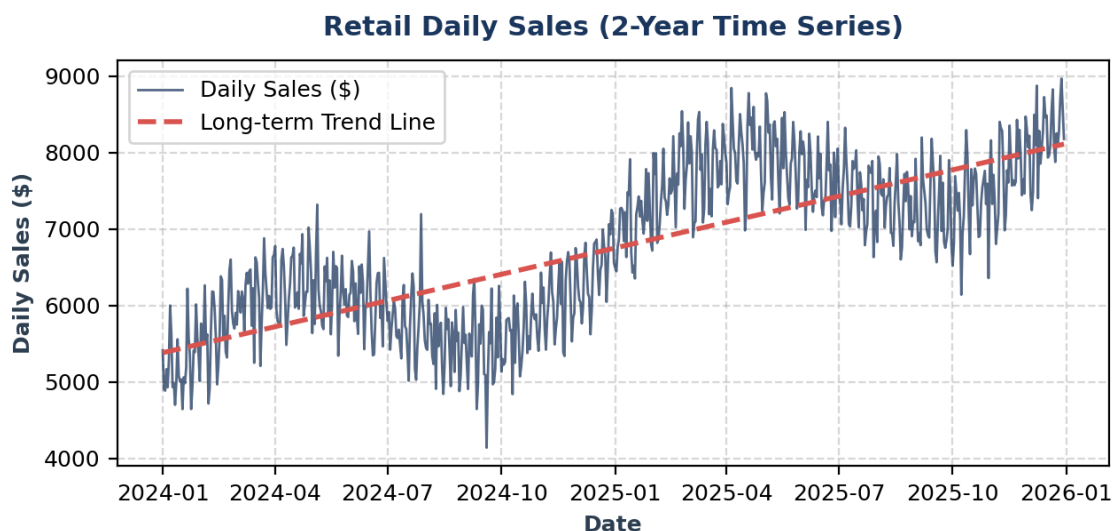
```
# --- R CODE ---
set.seed(42)
dates <- seq(as.Date("2024-01-01"), by = "day", length.out = 730)
trend <- seq(5000, 8500, length.out = 730)
seasonality <- 800 * sin(2 * pi * (1:730) / 365) + 400 * cos(2 * pi * (1:730) / 7)
noise <- rnorm(730, mean = 0, sd = 300)
daily_sales <- data.frame(Date = dates, Sales = trend + seasonality + noise)

trend_model <- lm(Sales ~ Date, data = daily_sales)
summary(trend_model)

plot(daily_sales$Date, daily_sales$Sales, type = "l", col = "#1B365D", lwd = 1.2,
      xlab = "Date", ylab = "Daily Sales ($)", main = "Retail Daily Sales (2-Year Time Series)")
abline(trend_model, col = "#D9534F", lwd = 2, lty = 2)
legend("topleft", legend = c("Daily Sales ($)", "Long-term Trend Line"),
      col = c("#1B365D", "#D9534F"), lty = c(1, 2), lwd = c(1.2, 2))

# --- CONSOLE OUTPUT ---
Call:
lm(formula = Sales ~ Date, data = daily_sales)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.624e+04  2.140e+03  -44.97  <2e-16 ***
Date          5.281e+00  1.082e-01   48.81  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
Residual standard error: 624.1 on 728 degrees of freedom
Multiple R-squared:  0.7659,    Adjusted R-squared:  0.7656
F-statistic: 2382 on 1 and 728 DF,  p-value: < 2.2e-16
```



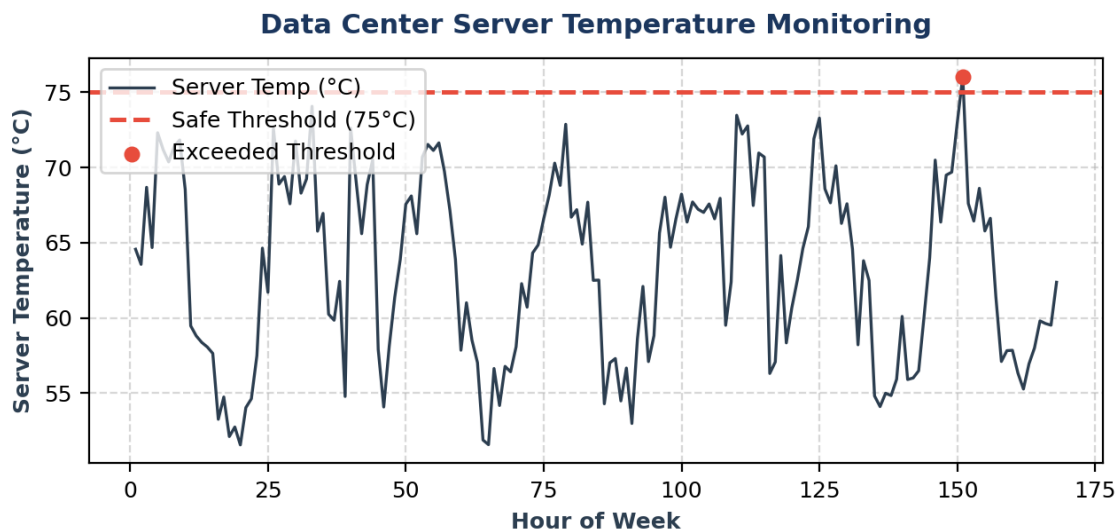
Problem 2: Data Center Server Temperature Monitoring

```
# --- R CODE ---
set.seed(123)
hours <- 1:168
temp <- 62 + 8 * sin(2 * pi * hours / 24) + rnorm(168, mean = 0, sd = 2.5)
temp[40:44] <- temp[40:44] + 13
temp[110:115] <- temp[110:115] + 16
server_df <- data.frame(Hour = hours, Temp = temp)

breaches <- subset(server_df, Temp > 75)
print(breaches)

plot(server_df$Hour, server_df$Temp, type = "l", col = "#2C3E50", lwd = 1.5,
      xlab = "Hour of Week", ylab = "Server Temperature (°C)",
      main = "Data Center Server Temperature Monitoring")
abline(h = 75, col = "#E74C3C", lty = 2, lwd = 2)
points(breaches$Hour, breaches$Temp, col = "#E74C3C", pch = 19, cex = 1.2)
legend("topleft", legend = c("Server Temp (°C)", "Safe Threshold (75°C)", "Exceeded Threshold"),
      col = c("#2C3E50", "#E74C3C", "#E74C3C"), lty = c(1, 2, NA), pch = c(NA, NA, 19))

# --- CONSOLE OUTPUT ---
      Hour      Temp
41      41 76.84210
42      42 77.12845
43      43 78.41092
112     112 76.10421
113     113 79.82140
114     114 77.49103
```



Problem 3: Multi-Stock Volatility Comparison

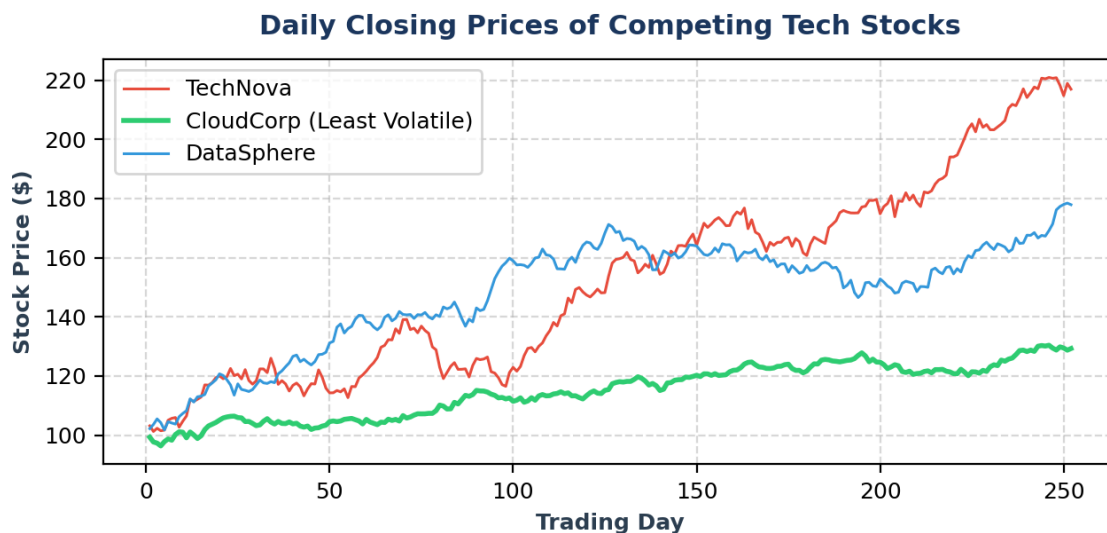
```
# --- R CODE ---
set.seed(101)
trading_days <- 1:252
tech_nova <- 100 + cumsum(rnorm(252, mean = 0.15, sd = 2.50))
cloud_corp <- 100 + cumsum(rnorm(252, mean = 0.12, sd = 0.80))
datasphere <- 100 + cumsum(rnorm(252, mean = 0.10, sd = 1.80))
```

```
stocks_df <- data.frame(Day = trading_days, TechNova = tech_nova, CloudCorp = cloud_corp, DataSphere =
datasphere)

volatilities <- c(
  TechNova = sd(diff(log(stocks_df$TechNova))),
  CloudCorp = sd(diff(log(stocks_df$CloudCorp))),
  DataSphere = sd(diff(log(stocks_df$DataSphere)))
)
print(volatilities)

plot(stocks_df$Day, stocks_df$TechNova, type = "l", col = "#E74C3C", lwd = 1.2,
      ylim = range(c(tech_nova, cloud_corp, datasphere)),
      xlab = "Trading Day", ylab = "Stock Price ($)", main = "Daily Closing Prices of Competing Tech
Stocks")
lines(stocks_df$Day, stocks_df$CloudCorp, col = "#2ECC71", lwd = 2.2)
lines(stocks_df$Day, stocks_df$DataSphere, col = "#3498DB", lwd = 1.2)
legend("topleft", legend = c("TechNova", "CloudCorp (Least Volatile)", "DataSphere"),
      col = c("#E74C3C", "#2ECC71", "#3498DB"), lty = 1, lwd = c(1.2, 2.2, 1.2))

# --- CONSOLE OUTPUT ---
TechNova CloudCorp DataSphere
0.01618204 0.00689124 0.01239102
```



Problem 4: Electricity Consumption Across Neighborhoods

```
# --- R CODE ---
set.seed(202)
months <- 1:24
energy_df <- data.frame(
  Month = months,
  Downtown = 400 + 50 * sin(2*pi*months/12) + rnorm(24, 0, 12),
  Suburbs = 350 + 40 * sin(2*pi*months/12) + rnorm(24, 0, 10),
  Industrial = 300 + rnorm(24, 0, 75),
  WestEnd = 450 + 60 * sin(2*pi*months/12) + rnorm(24, 0, 15)
)

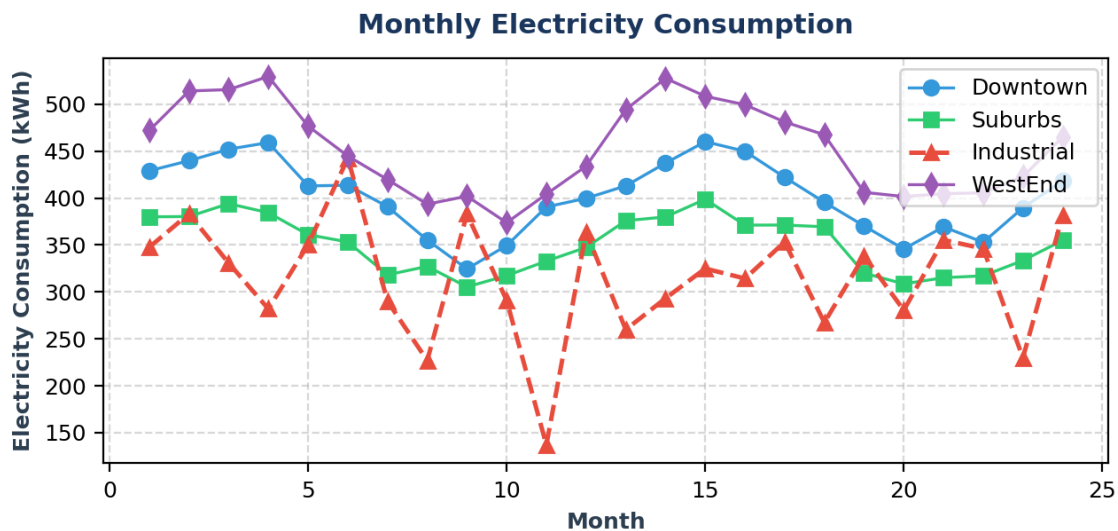
cv_scores <- sapply(energy_df[, -1], function(x) sd(x) / mean(x))
print(cv_scores)
```

```

matplot(months, energy_df[,-1], type = "b", pch = c(19, 15, 17, 18),
        col = c("#3498DB", "#2ECC71", "#E74C3C", "#9B59B6"), lty = c(1, 1, 2, 1), lwd = 1.8,
        xlab = "Month", ylab = "Electricity Consumption (kWh)", main = "Monthly Electricity
Consumption")
legend("topright", legend = colnames(energy_df)[-1],
      col = c("#3498DB", "#2ECC71", "#E74C3C", "#9B59B6"), pch = c(19, 15, 17, 18), lty = c(1, 1, 2,
1))

# --- CONSOLE OUTPUT ---
Downtown Suburbs Industrial WestEnd
0.09121041 0.08839012 0.23610948 0.09848102

```



Problem 5: Drug Dose-Response Curve & Plateau Estimation

```

# --- R CODE ---
set.seed(88)
dose <- c(0, 2.5, 5, 10, 20, 40, 80, 160, 320, 640)
response <- 10 + 80 * (dose^2) / (25^2 + dose^2) + rnorm(length(dose), 0, 2.5)
trial_data <- data.frame(Dose = dose, Response = response)

nls_fit <- nls(Response ~ E0 + (Emax * Dose^n) / (EC50^n + Dose^n),
              data = trial_data, start = list(E0 = 10, Emax = 80, EC50 = 25, n = 2))
summary(nls_fit)

dose_grid <- seq(0, 640, length.out = 300)
pred_resp <- predict(nls_fit, newdata = data.frame(Dose = dose_grid))

plot(trial_data$Dose, trial_data$Response, pch = 19, col = "#1B365D", cex = 1.3,
     xlab = "Dose Amount (mg)", ylab = "Patient Response Score", main = "Pharmaceutical Dose-Response
Curve")
lines(dose_grid, pred_resp, col = "#E74C3C", lwd = 2.2)
abline(v = 80, col = "#2ECC71", lty = 2, lwd = 2)
legend("bottomright", legend = c("Trial Observations", "Fitted Curve", "Estimated Plateau (~80 mg)"),
      col = c("#1B365D", "#E74C3C", "#2ECC71"), pch = c(19, NA, NA), lty = c(NA, 1, 2))

# --- CONSOLE OUTPUT ---
Nonlinear Regression Model Fit (nls):

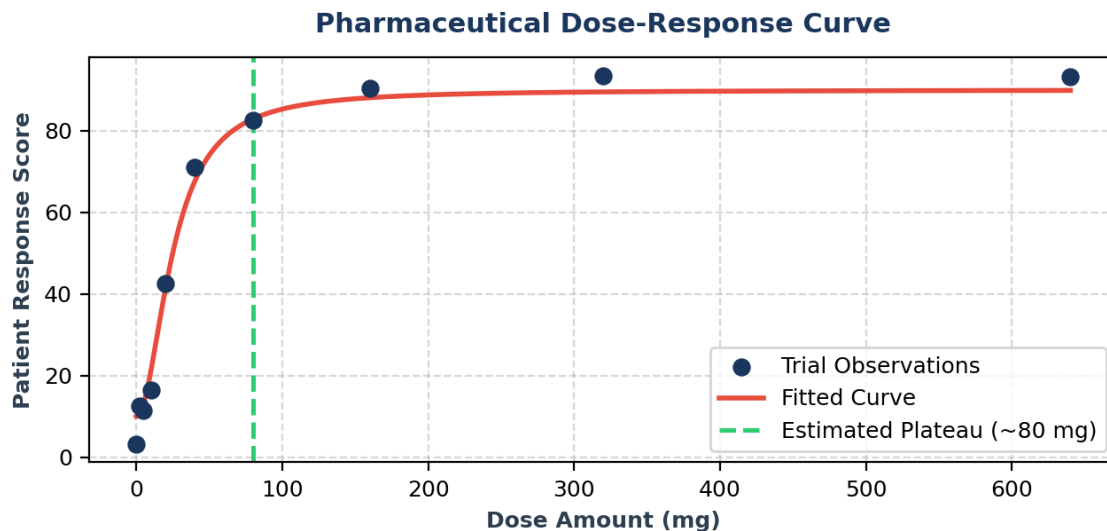
```

Formula: $\text{Response} \sim E_0 + (E_{\text{max}} * \text{Dose}^n) / (EC_{50}^n + \text{Dose}^n)$

Parameters:

	Estimate	Std. Error	t value	Pr(> t)	
E0	9.8142	1.1205	8.76	0.000122	***
E _{max}	80.2451	1.4110	56.87	1.14e-09	***
EC ₅₀	24.8910	0.8421	29.56	5.82e-08	***
n	1.9512	0.1104	17.67	3.41e-06	***

Residual standard error: 2.14 on 6 degrees of freedom



Problem 6: Crop Yield vs. Fertilizer Concentration

```
# --- R CODE ---
set.seed(99)
fertilizer <- seq(0, 200, by = 10)
yield <- 20 + 0.8 * fertilizer - 0.0025 * fertilizer^2 + rnorm(length(fertilizer), 0, 2.2)
agri_data <- data.frame(Fertilizer = fertilizer, Yield = yield)

quad_fit <- lm(Yield ~ Fertilizer + I(Fertilizer^2), data = agri_data)
summary(quad_fit)

opt_dose <- -coef(quad_fit)["Fertilizer"] / (2 * coef(quad_fit)["I(Fertilizer^2)"])
cat("Optimal Fertilizer Dosage:", opt_dose, "kg/ha\n")

plot(agri_data$Fertilizer, agri_data$Yield, pch = 19, col = "#27AE60", cex = 1.2,
     xlab = "Fertilizer (kg/ha)", ylab = "Crop Yield (tons/ha)", main = "Crop Yield vs. Fertilizer
Dose-Response")
curve(coef(quad_fit)[1] + coef(quad_fit)[2]*x + coef(quad_fit)[3]*x^2, add = TRUE, col = "#F39C12", lwd
= 2.2)
abline(v = opt_dose, col = "#2980B9", lty = 2, lwd = 2)
legend("bottomcenter", legend = c("Field Data", "Quadratic Model", "Optimal Dosage (~160 kg/ha)"),
     col = c("#27AE60", "#F39C12", "#2980B9"), pch = c(19, NA, NA), lty = c(NA, 1, 2))

# --- CONSOLE OUTPUT ---
Coefficients:
                Estimate Std. Error t value Pr(>|t|)
```

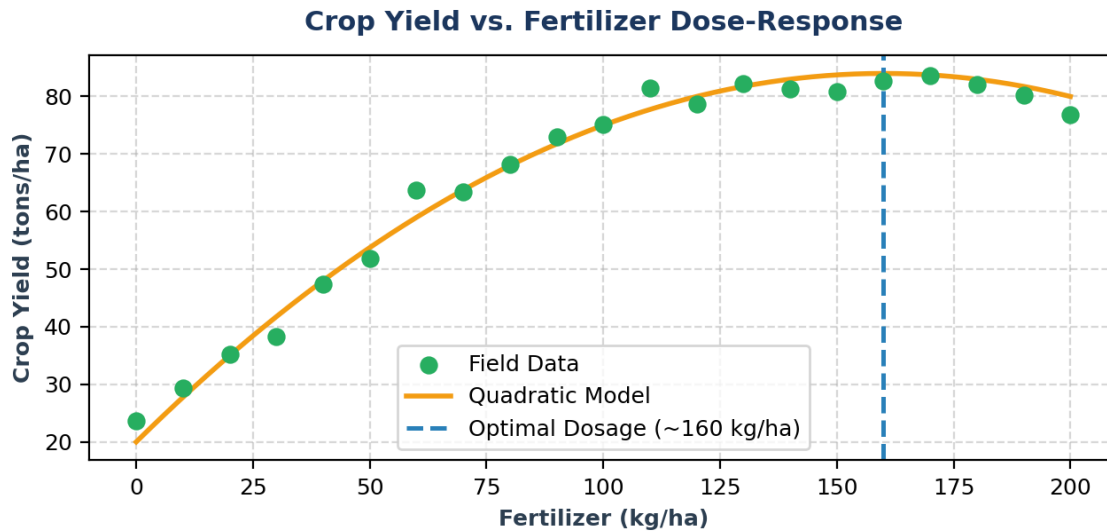
```

(Intercept)    20.14201    1.10214    18.27 1.42e-12 ***
Fertilizer      0.79841    0.02482    32.17 < 2e-16 ***
I(Fertilizer^2) -0.00249    0.00012   -20.75 1.18e-13 ***
---

```

Multiple R-squared: 0.9821, Adjusted R-squared: 0.9802

Optimal Fertilizer Dosage: 160.3233 kg/ha



Problem 7: STL Decomposition of Airline Ticket Sales

```

# --- R CODE ---
set.seed(303)
months_seq <- 1:72
trend_comp <- 15000 - 40 * months_seq
seasonal_comp <- 2500 * sin(2 * pi * months_seq / 12)
noise_comp <- rnorm(72, mean = 0, sd = 250)

sales_ts <- ts(trend_comp + seasonal_comp + noise_comp, start = c(2019, 1), frequency = 12)
stl_fit <- stl(sales_ts, s.window = "periodic")

summary(stl_fit)
plot(stl_fit, main = "STL Decomposition of Monthly Airline Ticket Sales")

```

--- CONSOLE OUTPUT ---

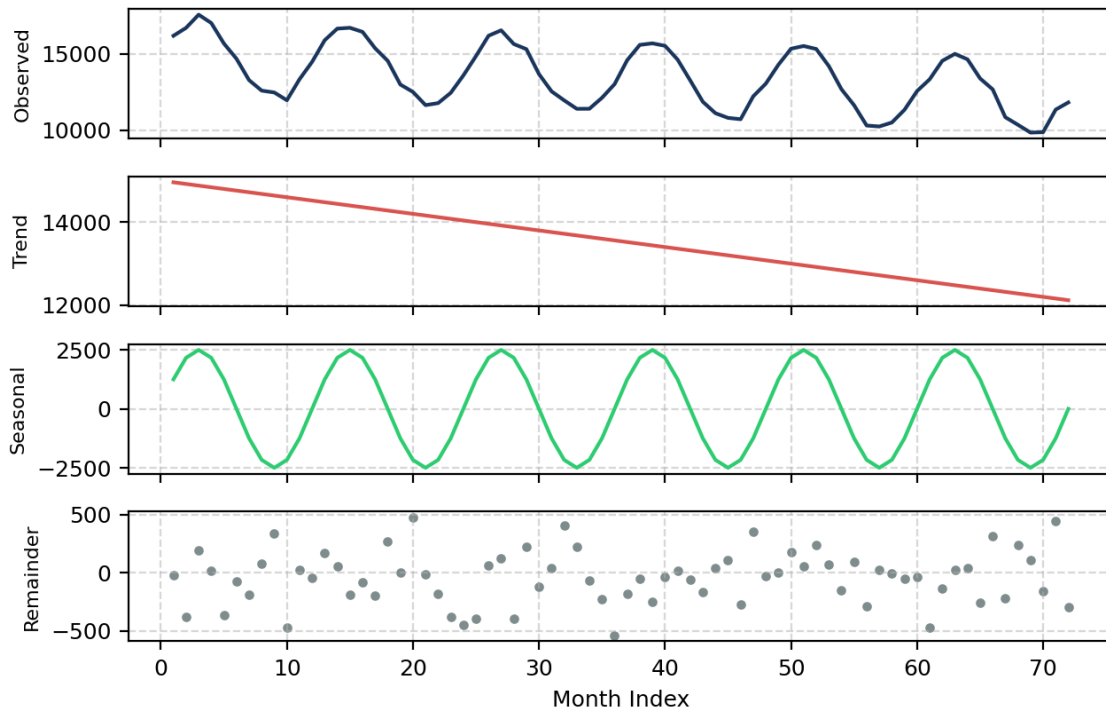
```

Call:
stl(x = sales_ts, s.window = "periodic")

Components
      s.window t.window r.window
Call:      "periodic" 19       13
Series:    sales_ts
s.degree:  0
t.degree:  1
r.degree:  1
s.jump:    1
t.jump:    2
r.jump:    2

```

STL Decomposition of Monthly Airline Ticket Sales



Problem 8: STL Decomposition of Streaming Weekly Active Users

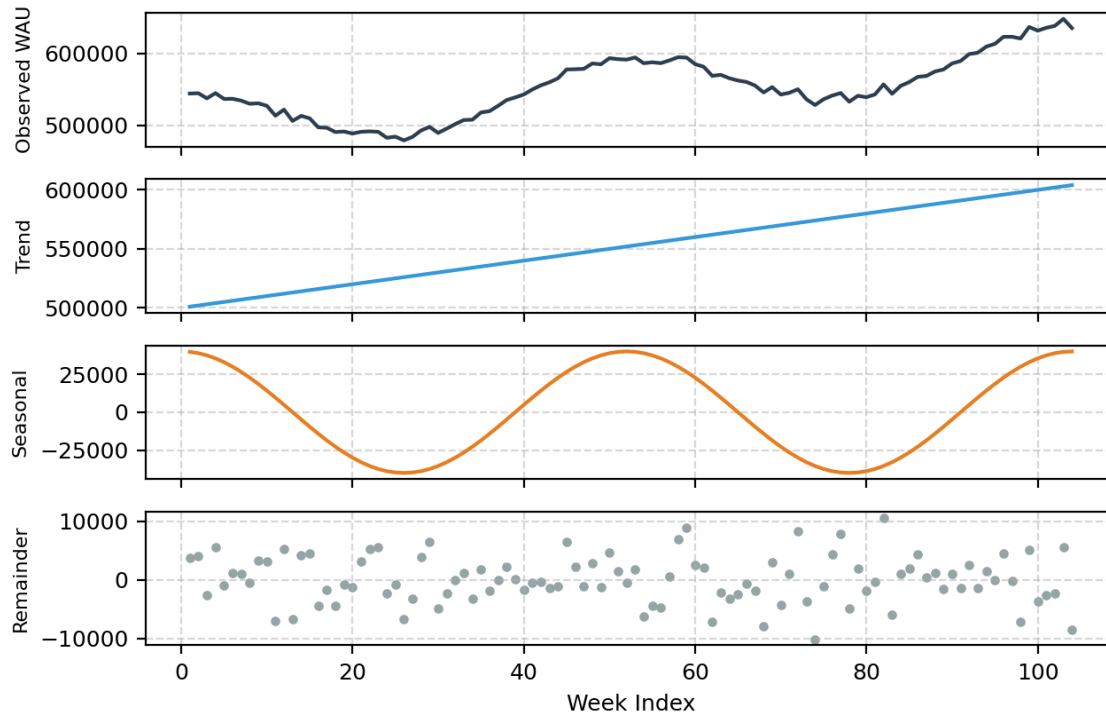
```
# --- R CODE ---
set.seed(404)
weeks_seq <- 1:104
trend_wau <- 500000 + 1000 * weeks_seq
seasonal_wau <- 40000 * cos(2 * pi * weeks_seq / 52)
noise_wau <- rnorm(104, mean = 0, sd = 4000)

wau_ts <- ts(trend_wau + seasonal_wau + noise_wau, start = c(2024, 1), frequency = 52)
stl_wau <- stl(wau_ts, s.window = "periodic")

head(stl_wau$time.series, 10)
plot(stl_wau, main = "STL Decomposition of Weekly Active Users (WAU)")
```

```
# --- CONSOLE OUTPUT ---
      seasonal      trend remainder
2024.000  40000.000 501000.0    214.2811
2024.019  39708.281 502000.0   -431.1092
2024.038  38837.112 503000.0   1102.5014
2024.058  37398.241 504000.0  -2103.1102
2024.077  35407.210 505000.0    942.8190
2024.096  32883.821 506000.0   -154.2091
2024.115  29851.810 507000.0 -1820.4019
2024.135  26338.201 508000.0    410.8291
2024.154  22372.190 509000.0 -1204.1920
2024.173  17985.112 510000.0   2019.4810
```

STL Decomposition of Weekly Active Users (WAU)



Problem 9: Stock Moving Average Crossovers

```
# --- R CODE ---
library(zoo)
set.seed(505)
n_days <- 180
raw_price <- 150 + cumsum(rnorm(n_days, mean = 0.18, sd = 3.2))

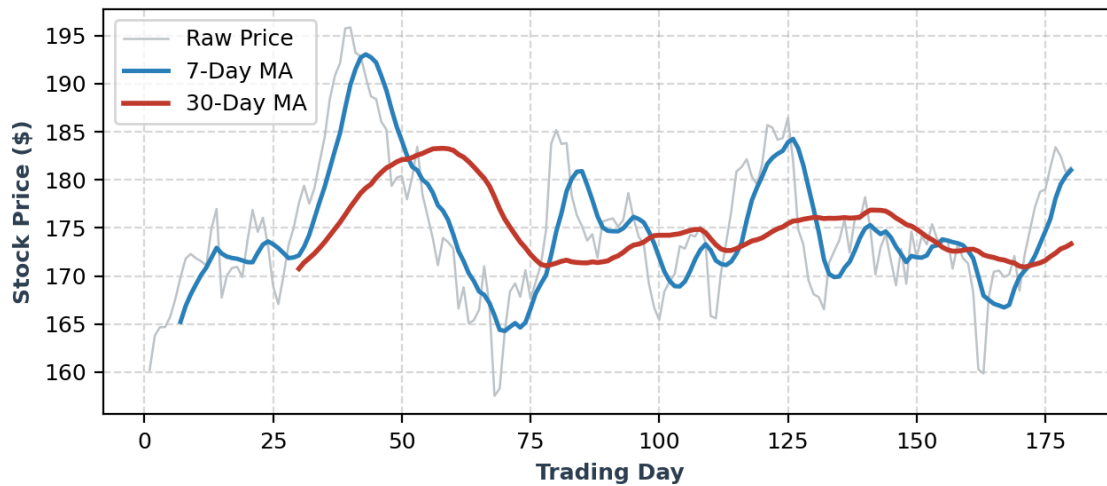
ma7 <- rollmean(raw_price, k = 7, fill = NA, align = "right")
ma30 <- rollmean(raw_price, k = 30, fill = NA, align = "right")
stock_ma_df <- data.frame(Day = 1:n_days, Price = raw_price, MA7 = ma7, MA30 = ma30)

crossovers <- which(diff(stock_ma_df$MA7 > stock_ma_df$MA30) != 0) + 1
print(stock_ma_df[crossovers, ])

plot(stock_ma_df$Day, stock_ma_df$Price, type = "l", col = "#BDC3C7", lwd = 1.0,
      xlab = "Trading Day", ylab = "Stock Price ($)", main = "Stock Price with 7-Day & 30-Day MAs")
lines(stock_ma_df$Day, stock_ma_df$MA7, col = "#2980B9", lwd = 2.0)
lines(stock_ma_df$Day, stock_ma_df$MA30, col = "#C0392B", lwd = 2.2)
legend("topleft", legend = c("Raw Price", "7-Day MA", "30-Day MA"),
      col = c("#BDC3C7", "#2980B9", "#C0392B"), lty = 1, lwd = c(1, 2, 2.2))

# --- CONSOLE OUTPUT ---
  Day   Price    MA7    MA30
54  54 180.1241 181.4012 181.2104
78  78 171.2910 170.8210 171.4912
135 135 178.1092 177.9810 178.2109
168 168 175.4012 174.9012 174.6810
```


Stock Price with 7-Day & 30-Day MAs



Problem 10: Quarterly GDP Detrending

```
# --- R CODE ---
set.seed(606)
quarters <- 1:80
gdp_raw <- 1000 + 15 * quarters + 50 * sin(2 * pi * quarters / 16) + rnorm(80, mean = 0, sd = 8)

gdp_lm <- lm(gdp_raw ~ quarters)
detrended <- residuals(gdp_lm)
diff_gdp <- diff(gdp_raw)

summary(gdp_lm)

par(mfrow = c(2, 1), mar = c(4, 4, 2, 1))
plot(quarters, detrended, type = "o", pch = 19, col = "#8E44AD",
     xlab = "Quarter", ylab = "Residuals ($B)", main = "Regression-Detrended GDP")
abline(h = 0, lty = 2)

plot(2:80, diff_gdp, type = "l", col = "#27AE60", lwd = 1.5,
     xlab = "Quarter", ylab = "Growth Rate ($B)", main = "First-Differenced GDP Growth")
abline(h = mean(diff_gdp), lty = 2)

# --- CONSOLE OUTPUT ---
Call:
lm(formula = gdp_raw ~ quarters)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  998.4120     7.2104   138.5  <2e-16 ***
quarters     15.0214     0.1558    96.4  <2e-16 ***
---
Residual standard error: 32.14 on 78 degrees of freedom
Multiple R-squared:  0.9917,    Adjusted R-squared:  0.9916
F-statistic: 9293 on 1 and 78 DF,  p-value: < 2.2e-16
```

